

Animal Kingdom — Rapid Revision Notes

PU-I Biology • Last-minute revision

Basis of Classification

Levels of organisation • Symmetry • Germ layers • Coelom • Segmentation • Notochord

Body Symmetry, Germ Layers and Coelom

Radial • Bilateral • Asymmetry • Diploblastic • Triploblastic • Coelomate/Pseudocoelomate/Acoelomate

Segmentation and Notochord

Metamerism • Chordates • Non-chordates

Phylum Porifera

Sponges • Cellular level • Canal system • Ostia/Spongocoel/Osculum

Phylum Coelenterata (Cnidaria)

Cnidoblasts • Polyp and medusa • Metagenesis

Phylum Ctenophora

Comb jellies • Bioluminescence • Comb plates

Phylum Platyhelminthes

Flatworms • Flame cells • Hermaphrodite

Phylum Aschelminthes

Round worms • Pseudocoelomates • Dioecious

Phylum Annelida

Segmented worms • Nephridia • Closed circulation

Phylum Arthropoda

Largest phylum • Jointed appendages • Open circulation • Malpighian tubules

Phylum Mollusca

Second largest phylum • Mantle cavity • Radula

Phylum Echinodermata

Spiny-bodied • Water vascular system • Marine

Phylum Hemichordata

Worm-like marine animals • Proboscis, collar, trunk

Phylum Chordata — General Features and Subphyla

Notochord • Dorsal nerve cord • Gill slits • Post-anal tail

Class Cyclostomata

Jawless • Ectoparasitic • Cartilaginous

Class Chondrichthyes

Cartilaginous fish • Placoid scales • Poikilothermous

Class Osteichthyes

Bony fish • Operculum • Air bladder

Class Amphibia

Water and land • Cloaca • Three-chambered heart

Class Reptilia

Creeping/crawling • Dry cornified skin • Oviparous

Class Aves

Wings and flight • Four-chambered heart • Homoiothermous

Class Mammalia

Mammary glands • Hair • Viviparous

Summary — Salient Features of Phyla

Quick comparison table